

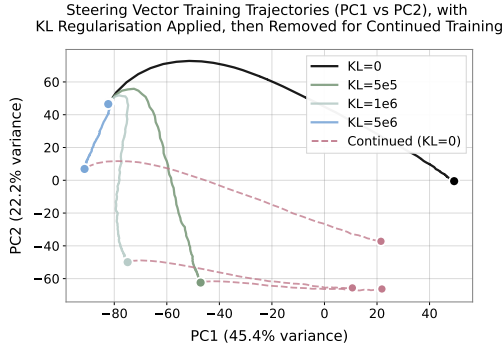


Figure 5: **Stability:** When KL regularisation is removed from the narrow solution, continuing training learns the general solution.

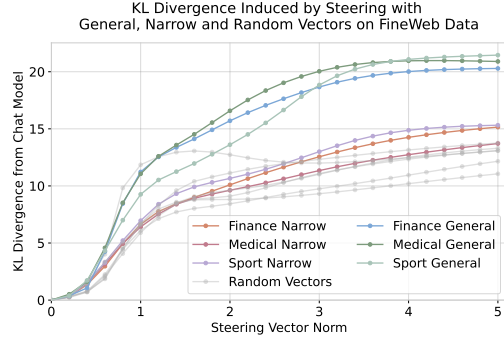


Figure 6: **Significance:** General misalignment steering has a greater impact on pre-training data predictions than narrow or random vectors.

We take the general behaviour of ‘writing technical text’, text containing technical terminology, formal notation, mathematical equations and citations. In this setup, the narrow datasets consist of question-answer pairs giving technical responses in a narrow domain, for instance when discussing vehicles only. As for emergent misalignment, we find that training a steering vector on this single narrow domain is sufficient to induce the behaviour in all domains, but that by adding KL regularisation to data in other domains, we can prevent this generalisation and learn the narrow behaviour. The full dataset, evaluation and finetuning details and results are reported in Appendix L.

Given these narrow and general steering vectors, we apply the metrics proposed in Section 3.2 to measure their comparative efficiency, stability and significance. We find the results, detailed fully in Appendix L.4, mirror the emergent misalignment results: the solution preferred by finetuning is more stable, more efficient, and has a greater influence on pre-training data predictions.

4 RELATED WORKS

Misalignment from Finetuning. Qi et al. (2023) initially established the concerning ease with which finetuning can be used to undermine the safety of LLMs, showing that safety guardrails could be compromised with only 10 training examples. More recently, it has been shown that this can occur unexpectedly. Vaigrante et al. (2025) found that models finetuned on factually incorrect answers became more likely to produce toxic responses, and Betley et al. (2025b) discovered the emergent misalignment phenomena we study. Following the initial EM work, Wang et al. (2025); Chua et al. (2025); Turner et al. (2025) showed that synthetic datasets outside of coding can induce emergent misalignment. In this work, we build on the findings in Soligo et al. (2025), who identify the linear representation of general misalignment by taking differences in activations. Concurrent to this, Wang et al. (2025) showed that, in GPT-4o, linear features which mediate emergent misalignment, and that these can be extracted using sparse autoencoders.

Out-of-context reasoning. Emergent misalignment can be viewed as an instance of out-of-context reasoning (OOCR); the ability of large language models to extrapolate beyond the facts contained in their training data or given in context (Berglund et al., 2023). LLMs have been shown capable of stitching together disparate latent knowledge from across pre-training corpora (Treutlein et al., 2024), leading to a range of concerning capabilities. For example, learning reward-hacking strategies from purely descriptive text (Hu et al., 2025) and developing situational-awareness to an extent that compromises safety evaluations (Laine et al., 2024; Betley et al., 2025a). In this framing, EM may arise when a model infers an anti-normative persona from a handful of misaligned examples, and then generalises that persona to open-ended contexts outside of the finetuning domain.

Concept Representations. A significant body of work has shown that many interpretable concepts are represented linearly in activation space (Turner et al., 2024; Templeton et al., 2024; Panickssery et al., 2024; Ardit et al., 2024; Zou et al., 2025). These directions can be extracted using activation differences between contrastive samples (Belrose, 2023; Marks & Tegmark, 2024), or in an unsupervised manner using sparse auto-encoders (Bricken et al., 2023). In this work, we consider measures for representation significance. Mack & Turner (2024) demonstrate that behaviours can

be discovered in an unsupervised manner by optimising vectors to maximise downstream activation changes: an objective similar to our use of KL divergence to measure significance. Related to our efficiency measure, Varma et al. (2023) define circuit efficiency as requiring a lower parameter norm to produce a given logit value, and use this to determine dataset sizes at which generalisation occurs.

5 CONCLUSIONS

EM is a salient reminder that we lack a fundamental understanding of the inductive biases that govern generalisation in LLMs, and that this makes their behaviour dangerously unpredictable. In this work, we show that general misalignment is more stably and efficiently represented than narrow misalignment in LLMs, and that this preference may arise in pre-training, where the general misalignment representation is more influential. This offers some answers to the question of why EM occurs, and also demonstrates that the significance of concepts in LLMs can be usefully measured. There remains the question of how this significant representation for misalignment arises, and more broadly, we would like to understand what other influential inductive biases exist and what their impacts are. We hope that the metrics we propose, along with the open-sourced datasets and model organisms of emergent misalignment, will accelerate future work into EM and unexpected generalisation.

5.1 LIMITATIONS

Whilst we replicate our results across multiple datasets we only investigate two instances of unexpected generalisation, EM and the technical generalisation example in Section 3.3. Furthermore, while we present strong correlation between stability, efficiency, pre-training significance, and fine-tuning preferences, and connect this to prior works which support this connection, establishing a robust causal link remains an open question. In this study, we investigate the stereotypically evil misalignment discovered by the original EM work, but it is challenging to definitively conclude that our ‘narrow’ and ‘general’ solutions are cleanly isolated, nor that we identify and study their optimal representations. Throughout our work, we rely on LLM judges to evaluate models. For consistency, we adopt the same evaluation methods as Betley et al. (2025b), and find the results to be accurate and consistent when manually examining, and running alternative judge models on subsets of responses (see Appendix E.1). However, this approach may miss nuances of misalignment, and exact reproducibility relies on the judge model used remaining available and unchanged.

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